

Nate Cirino

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Available May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Computer Systems, Computer Architecture, Computer Networks, Algorithms & Data Structures, Systems Security, Object-Oriented Design, Cybersecurity, Foundations of Software Engineering, Data Science

Honors: json-io OSS Contributor, Dean's List (all semesters), Tech Lead at Northeastern Electronic Racing, PawHacks

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Building a REST API that unifies 3 competing e-label suppliers (Vusion, Pricer, Solum) behind one interface, enabling dynamic price updates across 50,000+ shelf labels and streamlining new outlet launches
- Own 500+ front-of-house POS and e-label devices at 99.9% uptime with sub-3-minute average resolution across all four Wayfair retail brands (AllModern, Birch Lane, Joss & Main, Perigold)

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Overhauled JSON serialization across 20+ microservices as an open-source contributor to json-io, reducing average payload size 35% and cutting inter-service communication latency 28% under peak clinical query load
- Achieved 92% uptime for the Expert AI inference pipeline across 250K+ daily active clinicians
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing access for 10K+ clinicians with zero degradation during peak EHR synchronization windows

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Implemented a FAISS in-memory vector store with periodic re-indexing and AWS S3 Vectors for persistence, holding retrieval p95 under 150ms at peak while cutting vector storage cost roughly 70% vs. OpenSearch Serverless
- Built Sentinel, an AI SRE agent in production, cutting MTTR on P0/P1 incidents from 45 to 12 minutes and reducing MTTD 60% by correlating Prometheus, Tempo, and Loki telemetry via tool-use
- Re-orged 20+ engineers into product-aligned squads (Retrieval, Generation, Observability); now technical advisor

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Orchestrated 200+ Azure Data Factory pipelines across 10+ data sources at a 99.95% success rate, holding retry-induced latency overhead under 1% via exponential-backoff retry policies
- Reduced change failure rate from 15% to 2% with multi-stage Azure DevOps pipelines running automated unit, integration, and performance tests in staging to catch regressions before production

PROJECTS

Kepler | Rust, C++, Raft, LSM-tree, MVCC, Tokio

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and an LSM-tree storage engine, owning the design end to end from write path and compaction to failover testing
- Reduced API server latency 25% and held 99% uptime across 100 pods under write-heavy load; eliminated managed-etcd dependence via an MVCC layer, cutting infra cost \$2,500/month and raising TPS headroom 30%

NOAA AUV Sonar Detection | Python, PyTorch, AWS Neuron SDK, SageMaker

Apr. 2025 – Jul. 2025

- Optimized CNN inference to sub-50ms p99 latency on a single AWS Inferentia2 core for real-time onboard AUVs
- Trained CNN models on thermal sensor spectrograms to detect novel aquatic signatures, reaching 94.7% precision and 91.2% recall on a 10,000+ event test set while cutting false positives 75% over prior baselines

SKILLS & INTERESTS

Languages & Tools: Python, C++, C, Rust, Java, Go, Bash, Linux, SQL, MySQL, PostgreSQL, MongoDB, gRPC, GraphQL, TypeScript, JavaScript, Kotlin, Swift, Docker, Kubernetes, Terraform, Buildkite, AWS, Azure, GCP

Frameworks & Libraries: FastAPI, Flask, Spring Boot, Node.js, Express, React, Next.js, PyTorch, TensorFlow

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Interests: Avid Cleveland Browns fan, Weightlifting, Rock Climbing, BJJ, Running